

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 from sklearn.cluster import KMeans

```

```

1 df = pd.read_csv("Mall_Customers.csv")
2
3 df.head()

```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)	
0	1	Male	19	15	39	
1	2	Male	21	15	81	
2	3	Female	20	16	6	
3	4	Female	23	16	77	
4	5	Female	31	17	40	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```

1 print("Rows and Columns:", df.shape)
2
3 print("\nColumns:")
4 print(df.columns)
5
6 print("\nMissing Values:")
7 print(df.isnull().sum())

```

Rows and Columns: (200, 5)

Columns:

```
Index(['CustomerID', 'Gender', 'Age', 'Annual Income (k$)',
      'Spending Score (1-100)'],
      dtype='object')
```

Missing Values:

```
CustomerID      0
Gender          0
Age             0
Annual Income (k$)  0
Spending Score (1-100)  0
dtype: int64
```

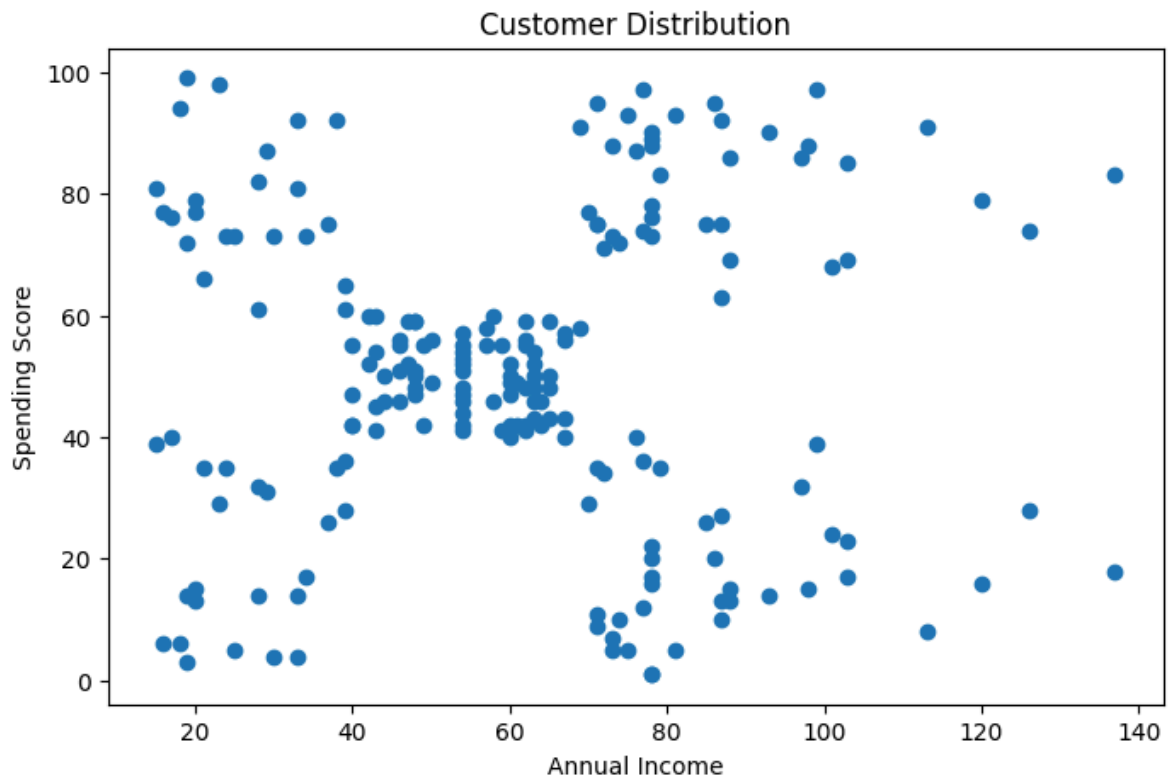
Annual Income vs Spending Score

```

1 plt.figure(figsize=(8,5))
2
3 plt.scatter(

```

```
4 df["Annual Income (k$)",  
5 df["Spending Score (1-100)"]  
6 )  
7  
8 plt.xlabel("Annual Income")  
9 plt.ylabel("Spending Score")  
10 plt.title("Customer Distribution")  
11  
12 plt.show()
```



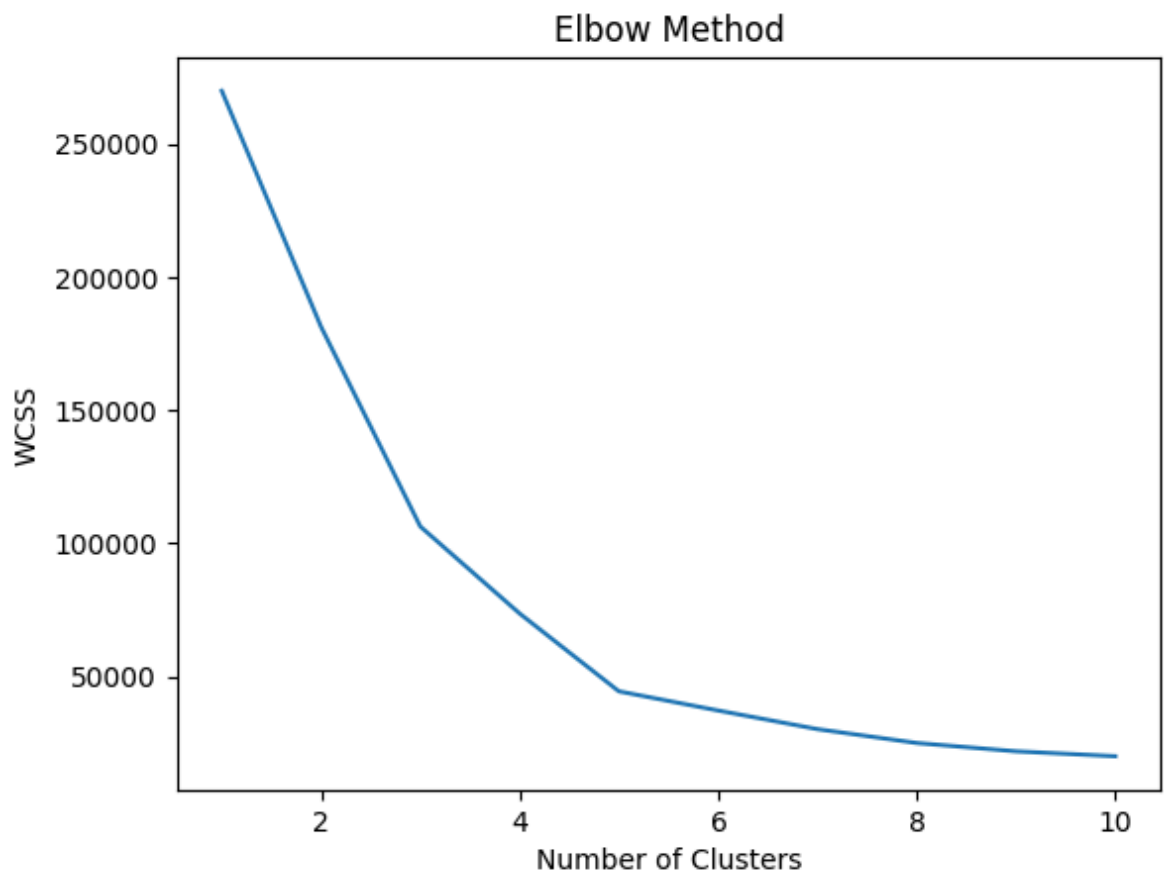
Select Features

```
1 X = df[[  
2     "Annual Income (k$)",  
3     "Spending Score (1-100)"  
4 ]]
```

Optimal Clusters

```
1 wcss = []  
2  
3 for i in range(1,11):  
4  
5     kmeans = KMeans(  
6         n_clusters=i,
```

```
7     random_state=42,  
8     n_init=10  
9 )  
10  
11 kmeans.fit(X)  
12  
13 wcss.append(kmeans.inertia_)  
14  
15 plt.plot(range(1,11), wcss)  
16  
17 plt.title("Elbow Method")  
18  
19 plt.xlabel("Number of Clusters")  
20  
21 plt.ylabel("WCSS")  
22
```



Applying K-Means

```
1 kmeans = KMeans(  
2     n_clusters=5,  
3     random_state=42,  
4     n_init=10  
5 )  
6  
7 df["Cluster"] = kmeans.fit_predict(X)
```

8

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)	Cluster	
0	1	Male	19	15	39	4	
1	2	Male	21	15	81	2	
2	3	Female	20	16	6	4	
3	4	Female	23	16	77	2	
4	5	Female	31	17	40	4	

Next steps:

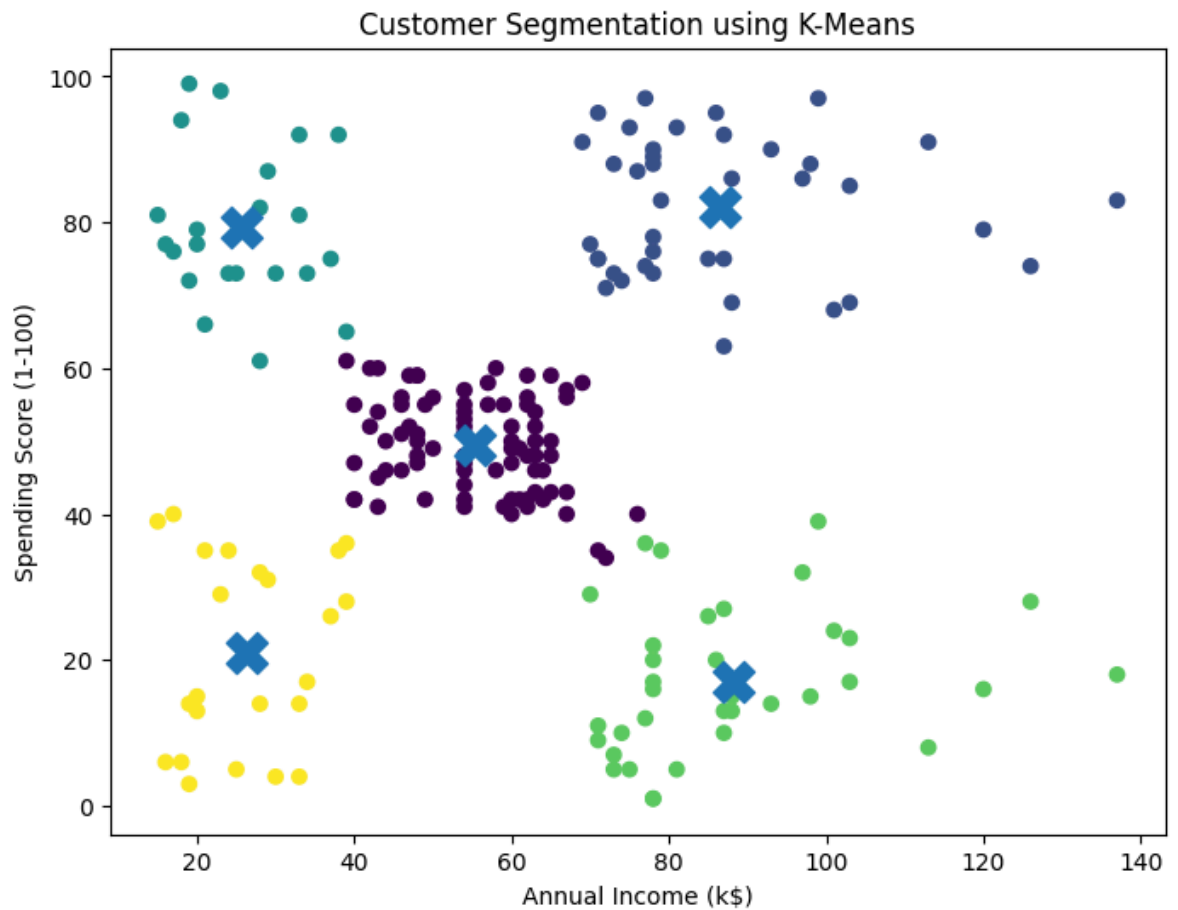
[Generate code with df](#)[New interactive sheet](#)

Visualize Customer Segments

```

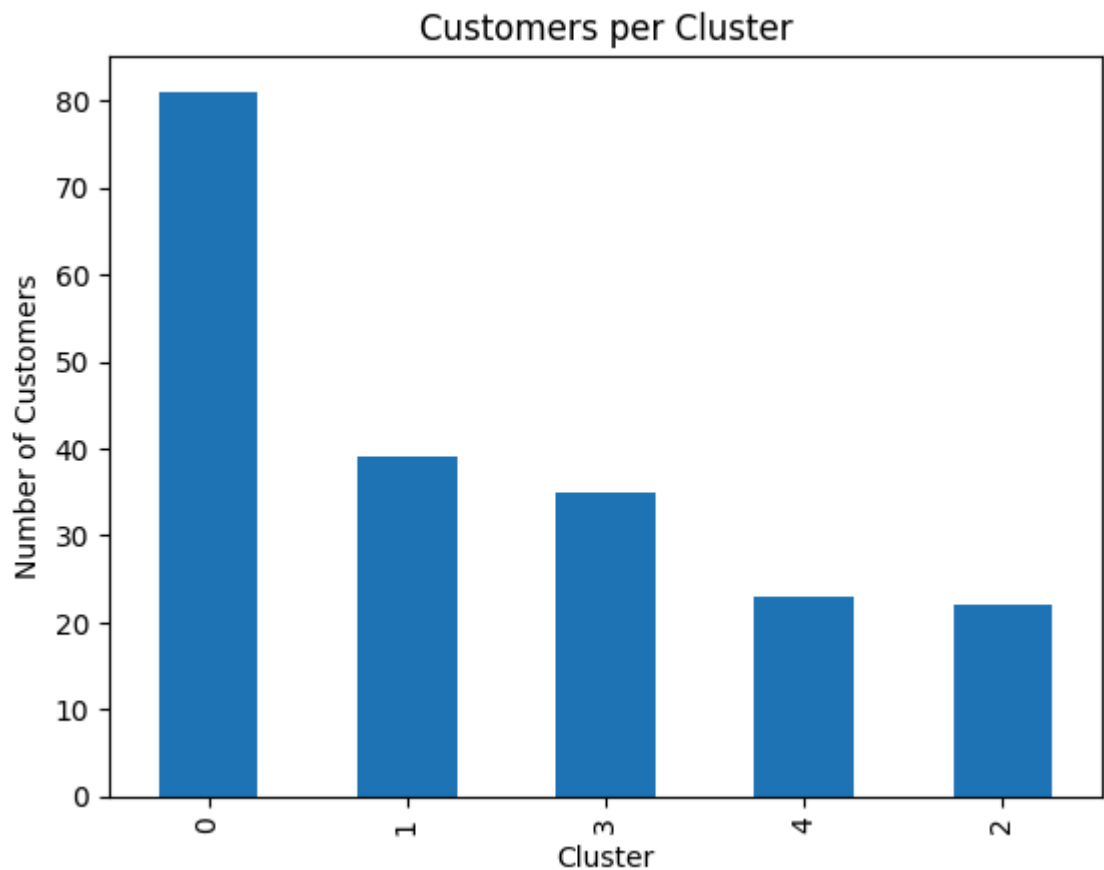
1 plt.figure(figsize=(8,6))
2
3 plt.scatter(
4     df["Annual Income (k$)"],
5     df["Spending Score (1-100)"],
6     c=df["Cluster"]
7 )
8
9 plt.scatter(
10     kmeans.cluster_centers[:,0],
11     kmeans.cluster_centers[:,1],
12     s=300,
13     marker="X"
14 )
15
16 plt.xlabel("Annual Income (k$)")
17 plt.ylabel("Spending Score (1-100)")
18 plt.title("Customer Segmentation using K-Means")
19
20 plt.show()

```



Cluster Distribution Chart

```
1 df["Cluster"].value_counts().plot(  
2     kind="bar"  
3 )  
4  
5 plt.title("Customers per Cluster")  
6 plt.xlabel("Cluster")  
7 plt.ylabel("Number of Customers")  
8  
9 plt.show()
```



Counting Customers in Each Cluster

```
1 print(df["Cluster"].value_counts())
```

```
Cluster
0      81
1      39
3      35
4      23
2      22
Name: count, dtype: int64
```

Cluster Centers

```
1 centers = pd.DataFrame(
2     kmeans.cluster_centers_,
3     columns=[
4         "Annual Income",
5         "Spending Score"
6     ]
7 )
8
9 print(centers)
```

```
   Annual Income  Spending Score
0    55.296296    49.518519
1    86.538462    82.128205
```

2	25.727273	79.363636
3	88.200000	17.114286
4	26.304348	20.913043

Cluster	Annual Income	Spending Score	Segment
1	86.54	82.13	High Value Customers
3	88.20	17.11	High Income, Low Spending
2	25.73	79.36	Low Income, High Spending
4	26.30	20.91	Low Value Customers
0	55.30	49.52	Average Customers

Key Findings

1. Customers were successfully segmented into 5 distinct groups using K-Means clustering.
2. Cluster 1 represents High Value Customers with high income and high spending behavior.
3. Cluster 3 contains customers with high income but low spending, indicating opportunities for targeted marketing campaigns.
4. Cluster 2 includes customers with lower income but high spending behavior, showing strong engagement with products and services.
5. Cluster 4 consists of low-income and low-spending customers who contribute the least revenue.
6. Cluster 0 represents average customers with moderate income and spending levels.
7. Customer segmentation can help businesses personalize marketing strategies and improve customer retention.

Business Recommendations

1. Offer loyalty rewards and premium services to High Value Customers.
2. Create targeted promotions for High Income, Low Spending customers to increase purchases.
3. Develop personalized marketing campaigns for highly engaged customers.
4. Allocate marketing budgets based on customer segment value.